



Document 4: SDLC Model

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1. Selected SDLC Model: Agile (Iterative)

The LearnHub project adopts the Agile (Iterative) Software Development Life Cycle model. Given the nature of the project – a single-developer mobile application with evolving requirements and incremental feature delivery – Agile provides the most suitable framework for managing development effectively.

1.1 Justification for Choosing Agile

Reason	Explanation
Incremental Delivery	Features (onboarding, search, enrollment, progress) were built and tested independently in iterations.
Flexibility	Requirements could be refined at any iteration without disrupting the overall plan.
Continuous Testing	Each feature was tested as it was developed, reducing the risk of large-scale bugs at the end.
Solo Developer Fit	Agile's lightweight process adapts well to a single developer managing all roles.
Short Feedback Loops	Working features were available for review after each iteration, enabling early feedback.

1.2 Agile Phases Applied to LearnHub

Sprint	Phase	Activities	Deliverable
1	Planning	Define requirements, user stories, tech stack, project timeline.	Project plan, user story backlog
2	Design	UI/UX wireframes, architecture design, use case diagrams.	Mockups, system diagrams
3	Development I	Build splash/onboarding, home screen, search & filter.	Functional home + search screens
4	Development II	Course detail, enrollment system, SharedPreferences persistence.	Enrollment + detail screens
5	Development III	Progress tracking, My Courses screen, Profile & Settings, dark mode.	Full app functionality
6	Testing	Unit tests, widget tests, integration testing, bug fixes.	Test reports, stable build
7	Deployment	Final build, documentation,	Final APK + documentation

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1.3 SDLC Diagram (Agile Iteration Cycle)

The Agile SDLC for LearnHub follows a cyclical process across each sprint:



→ Each arrow above represents a sprint cycle. After Review, the team returns to Plan for the next sprint.

1.4 Comparison with Other SDLC Models

Model	Why Not Chosen	LearnHub Consideration
Waterfall	Rigid sequential phases – no flexibility for iterative changes.	LearnHub features evolved during development.
Spiral	Focused on risk management – adds unnecessary overhead for small projects.	Low-risk, small-scale solo project.
V-Model	Testing only after full development – delays feedback.	Continuous testing was required per feature.
RAD	Requires multiple teams and heavy prototyping resources.	Solo developer with limited resources.